Experiment 1: Deploy and Manage an EC2 Instance

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Objective

- Launch and connect to an Amazon EC2 instance
- Configure a web server on the instance

Introduction

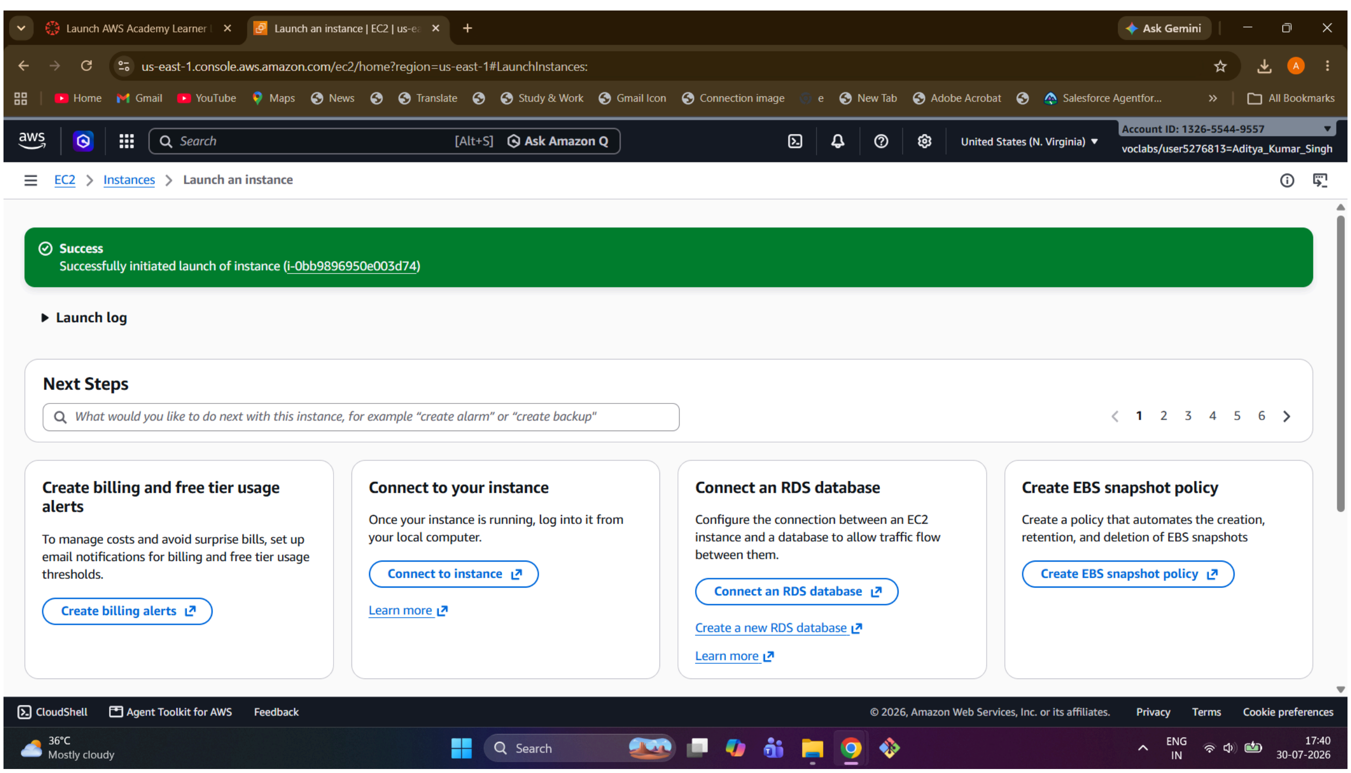
Amazon Elastic Compute Cloud (EC2) is a core AWS service that provides resizable virtual servers, known as instances, in the cloud. It allows users to quickly provision compute capacity without investing in physical hardware, and offers full control over the operating system, networking, and installed software. This experiment demonstrates the end-to-end process of launching an EC2 instance, securely connecting to it, and deploying a functional Apache web server on it.

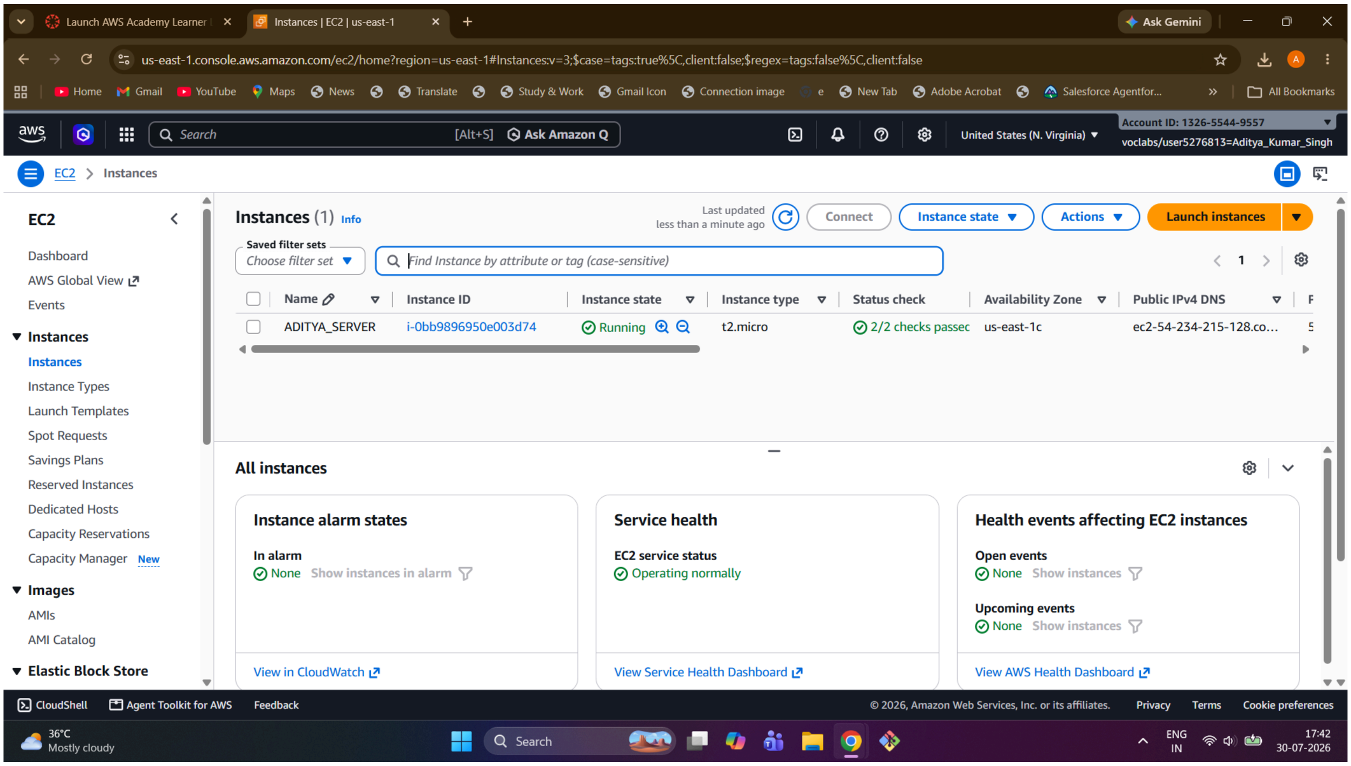
Procedure

1. Launch EC2 Instance

- Launched a t2.micro instance using the Amazon Linux 2 AMI.
- Configured a security group to allow inbound SSH (port 22) and HTTP (port 80) traffic.
- Created and downloaded the key pair (.pem file) required for secure SSH access.

EC2InstanceCreation.mp4





2. Connect to Instance

- Connected to the running instance from a local terminal using SSH and the downloaded key pair.
- Command used:

```
ssh -i my-ec2-key.pem ec2-user@<Public-IP>
```

- Verified successful login to the instance's shell prompt.

  EC2Connection.mp4

3. Configure Web Server

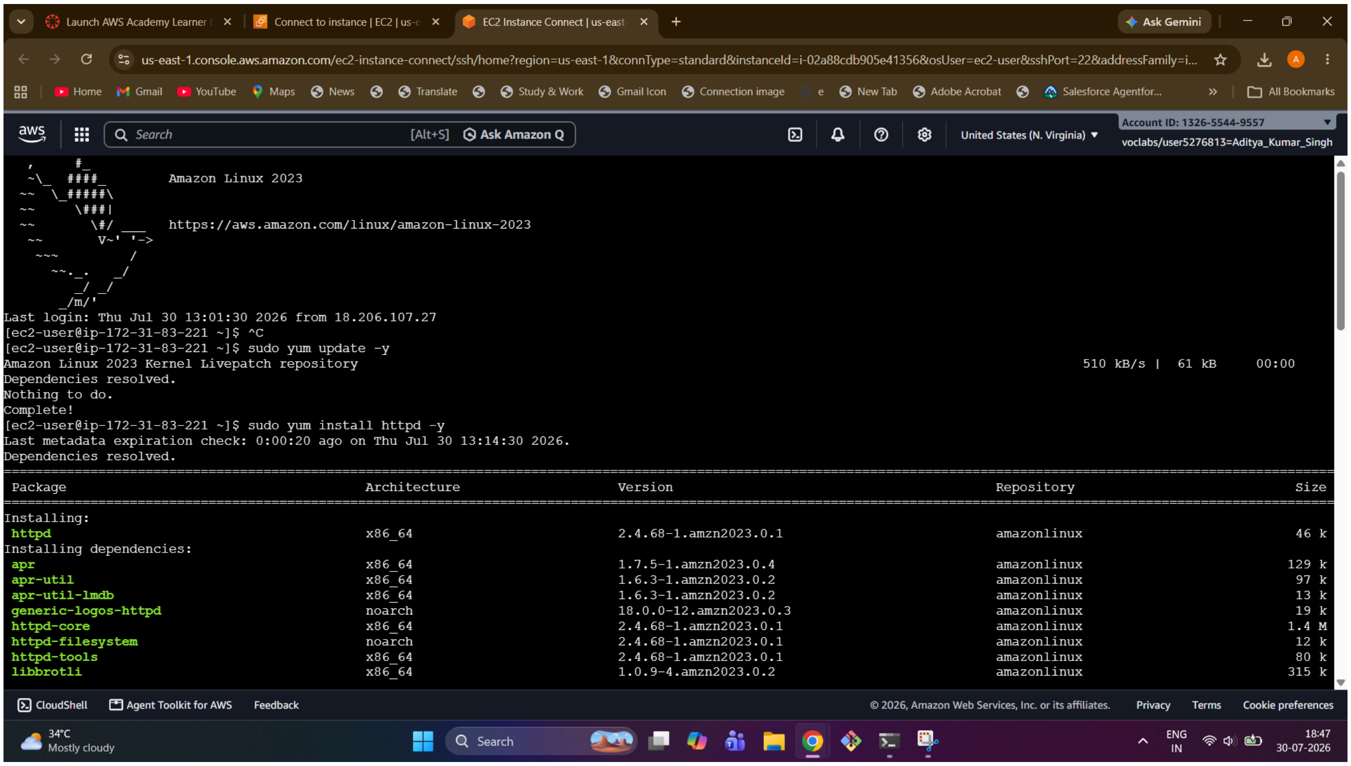
- Updated the system packages and installed the Apache HTTP server using the following commands:

```
sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd
```

- Started the httpd service and enabled it to launch automatically on instance reboot.
- Verified the setup by navigating to the instance's Public IP in a web browser, which displayed the default Apache test page.

```
ec2-user@ip-172-31-25-198:~

Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\BITTU>cd Downloads

C:\Users\BITTU\Downloads> cd my-key-pair.pem
The directory name is invalid.

C:\Users\BITTU\Downloads>ssh -i demokey:pem ec2-user@54.224.84.4
The authenticity of host '54.224.84.4 (54.224.84.4)' can't be established.
ED25519 key fingerprint is SHA256:Qscn7p8fTjv3z8hP5eZL6N9V4xk4y2b8nZ8x7y6w5v4U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.224.84.4' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
[ec2-user@ip-172-31-25-198 ~]$ |
```

```
ec2-user@ip-172-31-25-198:~

  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch        httpd-2.4.68-1.amzn2023.0.1.x86_64
  httpd-core-2.4.68-1.amzn2023.0.1.x86_64                  httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
  httpd-tools-2.4.68-1.amzn2023.0.1.x86_64                 libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch                     mod_http2-2.0.42-1.amzn2023.0.1.x86_64
  mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-25-198 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-25-198 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Sun 2026-08-02 09:00:39 UTC; 46s ago
       Docs: man:httpd.service(8)
   Main PID: 19917 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (Limit: 1059)
     Memory: 13.7M
        CPU: 122ms
     CGroup: /system.slice/httpd.service
             ├─19917 /usr/sbin/httpd -DFOREGROUND
             ├─20027 /usr/sbin/httpd -DFOREGROUND
             ├─20032 /usr/sbin/httpd -DFOREGROUND
             ├─20034 /usr/sbin/httpd -DFOREGROUND
             └─20082 /usr/sbin/httpd -DFOREGROUND

Aug 02 09:00:39 ip-172-31-25-198.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 02 09:00:39 ip-172-31-25-198.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Aug 02 09:00:39 ip-172-31-25-198.ec2.internal httpd[19917]: Server configured, listening on: port 80
[ec2-user@ip-172-31-25-198 ~]$
```


Launch AWS Academy Learner
Connect to instance | EC2 | us-e
EC2 Instance Connect | us-east
Ask Gemini
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-02a88cdb905e41356&osUser=ec2-user&sshPort=22&addressFamily=i...
Home Gmail YouTube Maps News Translate Study & Work Gmail Icon Connection image e New Tab Adobe Acrobat Salesforce Agentfor... All Bookmarks
Search [Alt+S] Ask Amazon Q
United States (N. Virginia)
Account ID: 1326-5544-9557
voclabs/user5276813=Aditya_Kumar_Singh
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Thu Jul 30 13:01:30 2026 from 18.206.107.27
[ec2-user@ip-172-31-83-221 ~]$ ^C
[ec2-user@ip-172-31-83-221 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository 510 kB/s | 61 kB 00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-83-221 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:00:20 ago on Thu Jul 30 13:14:30 2026.
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
httpd x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 46 k
Installing dependencies:
apr x86_64 1.7.5-1.amzn2023.0.4 amazonlinux 129 k
apr-util x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 97 k
apr-util-lmdb x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 13 k
generic-logos-httpd noarch 18.0.0-12.amzn2023.0.3 amazonlinux 19 k
httpd-core x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 1.4 M
httpd-filesystem noarch 2.4.68-1.amzn2023.0.1 amazonlinux 12 k
httpd-tools x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 80 k
libbrotli x86_64 1.0.9-4.amzn2023.0.2 amazonlinux 315 k
CloudShell Agent Toolkit for AWS Feedback
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34°C Mostly cloudy
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ENG IN
18:47 30-07-2026